CHAPTER-1

INTRODUCTION

Image Processing is the process or study that uses all varieties of techniques for manipulation to be used on digital images. Manipulation involves enhancement, analysis, and extracting meaningful information from those images. Core or mainly, an image is defined to be a two-dimensional function which gives a certain pixel either by intensity or colour. On the other hand, application of computer algorithms In relation to these digital images will denote Digital Image Processing.

1.1 THERE ARE PRIMARILY TWO TYPES OF IMAGE PROCESSING.

- Spatial Domain Processing: This directly manipulates the pixel values of an image. Filtering, noise reduction, and image enhancement are all examples of this technique.
- Frequency Domain Processing: Here, the image is transformed from the spatial domain to the frequency domain, in which case it is represented in terms of its frequency components. Filtering, for example, and image compression are often done in the frequency domain.

1.2 METHODS OF IMAGE PROCESSING

A wide range of methods and techniques are employed in image processing, including:

- Image Acquisition: Capturing images using digital cameras, scanners, or medical imaging devices.
- Image Enhancement: Improving the visual quality of an image, such as increasing contrast or sharpening edges.
- Image Restoration: Removing noise and artifacts from an image to restore its original quality.
- Image Segmentation: Dividing an image into meaningful regions or objects.
- Image Compression: It reduces the size of an image to facilitate storage and transmission.
- Image Analysis: It is extracting quantitative information from an image, such as shape, texture, and colour features.

1.3 THE IMPERATIVE OF IMAGE SECURITY: ENCRYPTION AND DECRYPTION

Since such information, though digital, represents a highly sensitive visual that people desire to be protected from leakage to unknown people, image encryption forms one of the indispensable techniques. In today's world of information and especially since this is highly graphical in nature, such pictures containing personal medical images or personal confidential documents or just family photographs have to remain encrypted in their unintelligible format against such malicious activities by certain unscrupulous minds of unwanted individuals.

1.3.1 Encryption Procedure

Image encryption is the process where cryptographic algorithms are applied to the original image data in such a way that it cannot be recognized. This usually generates a cryptographic key that will be used to encrypt and decrypt the image. Usually, strong encryption algorithms such as advanced encryption standard (AES) or data encryption standard (DES) are used for the encryption of the image.

1.3.2 Decryption Process

Image decryption is the reverse process of encryption, where the encrypted image is transformed back into its original form. This requires the correct decryption key to unlock the encrypted data. The decryption process involves the application of the same cryptographic algorithm used for encryption but with the decryption key.

Chaotic Approach: Logistic Map-Based Image Encryption

One of the more innovative approaches to image encryption is through the power of chaos theory. Chaotic systems are defined by their sensitivity to initial conditions and their lack of predictability. A classic example of a chaotic system is the logistic map-a simple nonlinear dynamical system. It is possible to exploit the inherent randomness of chaotic sequences to produce powerful encryption keys that are hard to crack.

RATIONALE

Healthcare has increasingly utilized digital technologies. As a result, digital medical images have increased in use. These images carry sensitive patient information and are transmitted and

stored electronically. For these images, there is a need for secure measures that ensure confidentiality and integrity.

Why Logistic Map-Based Encryption?

- Strong Encryption: Chaotic systems, like the logistic map, are characterized by sensitive dependence on initial conditions as well as complex dynamic behaviour. Therefore, they can be applied for encrypting keys that cannot be cracked or predicted.
- High Security: In general, logistic map-based encryption may generate a high level of security where the pixel values of an image scramble in a manner that seems complex and unpredictable.
- Efficient Implementation: The logistic map is a simple mathematical function. It is therefore computationally efficient to implement.
- Adaptability: The parameters of the logistic map can be adjusted to achieve different levels of security. This makes it adaptable to various security requirements.

Specific Benefits for Medical Images:

- Patient Privacy: Sensitive patient information can be protected by encrypting medical images before transmission or storage.
- Data Integrity: Medical images integrity can be ensured by preventing unauthorized modifications or tampering.
- Compliance with regulations: Observing data privacy regulations like HIPAA by ensuring the implementation of strong security measures.
- Telemedicine Security: Security of transmitting medical images during telemedicine sessions.
- Secure Cloud Storage: Protection of medical images that are stored in cloud systems from unauthorized access and breaches.

OBJECTIVES

- Investigate the use of the logistic map to generate chaotic sequences for encryption keys.
- Design an efficient image encryption algorithm based on the chaotic behaviour of the logistic map to scramble pixel values in medical images.

- Evaluate the security performance of the proposed encryption method, including key sensitivity tests, statistical analysis, sensitivity against differential attacks, and key space usage analysis.
- Determine the computational complexity of the proposed encryption and decryption algorithms.
- The practical ability of the proposed system will be illustrated by means of encryption and decryption of various types of medical images.
- Performance comparison of the proposed system with the existing image encryption techniques.

1.4 APPLICATIONS

- Secure Telemedicine:
  - Patient privacy while conducting telemedicine consultations is protected through the encryption of medical images being transmitted over networks.
  - The integrity of medical images to prevent tampering or unauthorized modifications.
- Secure Cloud Storage:
  - Encrypting medical images before uploading to the cloud-based storage system.
  - Protect patient confidentiality by encrypting medical images on local servers.
- Medical Image Archives:
  - Ensuring the long-term security of medical image archives by encrypting them to prevent unauthorized access.
  - Protecting medical images from natural disasters or cyberattacks by encrypting them before backup.
- Medical Research:
  - Securely sharing medical images with researchers and collaborators while maintaining patient privacy.
  - Protecting intellectual property rights associated with medical image data and research findings.
- Law Enforcement and Forensics:
  - Ensuring the integrity of forensic medical images by encrypting them to prevent tampering.

CHAPTER-2

LITERATURE SURVEY

2.1 CHAOTIC SYSTEMS AND IMAGE ENCRYPTION

2.1.1 Reference: [Gao, X., Mou, J., Xiong, L., Sha, Y., Yan, H. and Cao, Y., 2022]

Summary: This paper delves into the fundamental principles of chaotic systems and their application in the field of cryptography. It discusses the properties of chaotic maps, such as sensitivity to initial conditions, period-doubling bifurcations, and ergodicity, which make them suitable for generating cryptographic keys. The paper highlights the advantages of using chaotic systems for image encryption, including their ability to produce complex and unpredictable encryption keys, making it difficult for attackers to break the encryption.

2.2 LOGISTIC MAP-BASED IMAGE ENCRYPTION TECHNIQUES

2.2.1 Reference: [Çelik, H. and Doğan, N., 2024]

Abstract

This paper provides a novel encryption technique at pixel-level using logistic map. In the proposed approach, scrambling pixel values were done through the chaotic sequence generated using logistic map. The performance in terms of security is tested with regard to key sensitivity, statistical analysis, and resistance to differential attacks in this paper. Thus, it results in demonstrating effectiveness of this proposed approach.

2.2.2 Reference: [Arif, J., Khan, M.A., Ghaleb, B., Ahmad, J., Munir, A., Rashid, U. and Al-Dubai, A.Y., 2022]

Summary: This paper proposes the use of a block-level encryption by combining the logistic map with an encryption algorithm based on DNA. The image is fragmented into blocks and then separately encrypted using the DNA sequences obtained from the chaotic series. The authors further reveal that the new proposed system has enhanced security and higher robustness as compared with traditional encryption systems.

2.2.3 Ref: [Xiang, H. and Liu, L., 2020]

Summary: The presented paper introduces a hybrid encryption approach that combines the logistic map with a cellular automata-based encryption method. Logistic map chaotic sequence drives cellular automata evolution with significantly improved security. In the paper, security performance evaluation of the suggested method compared to other modern methods are presented.

There are a number of techniques that have been proposed for encrypting medical images using the logistic map. These techniques can be categorized into two broad approaches as follows:

i. Pixel-Level Encryption:

In this approach, each pixel of the image is encrypted independently using the chaotic sequence that is generated by the logistic map. There are many techniques, such as substitution and permutation, that may be used to scramble the pixel values.

For instance, a block-level encryption scheme has been proposed by [*Yue Hu, Ruyue Tian*, 2013-2024] using the logistic map to generate a random permutation sequence for pixel-level encryption.

ii. Block-Level Encryption:

Block-level encryption is also presented by dividing the image into smaller blocks and then independently encrypting each of those blocks. The whole process may be even further enhanced in terms of security and efficiency particularly for large images.

A block-level encryption scheme combining the logistic map with a DNA-based encryption technique was presented by [*Bowen Zhang*, 2023].

2.3 SECURITY ANALYSIS OF LOGISTIC MAP-BASED ENCRYPTION

2.3.1 Reference: [Rahman, Z., Yi, X., Khalil, I. and Sumi, M.,2021]

Abstract:

In this paper, the overall security analysis of logistic map-based encryption schemes is discussed. The authors describe and analyse various attack techniques against statistical attacks, differential attacks and brute-force attacks that happen to be applicable for this

type of encryption, using logistic maps. Importance for choosing appropriate parameters on logistic maps is also presented so as to enhance security.

2.3.2 Reference: [Wang, J., Han, K., Fan, S., Zhang, Y., Tan, H., Jeon, G., Pang, Y. and Lin, J., 2020]

Summary: This paper presents a new method of improvement in the security of encryption using logistic maps with dynamic key scheduling. The new mechanism will make sure the encryption key is changing during each iteration of the cipher, making it hard to break. This paper describes the efficiency of the approach in enhancing the security offered by logistic map-based encryption.

2.4 LOGISTIC MAP-BASED IMAGE ENCRYPTION IN MEDICAL IMAGING

2.4.1 Reference: [Kumar, M. and Gupta, P., 2021]

Abstract: This paper discusses the application of logistic map-based encryption in telemedicine. The authors proposed a secure transmission protocol that uses the logistic map to encrypt medical images before their transmission. The paper analysed the security and performance of the proposed protocol, showing its effectiveness in protecting patient privacy during remote consultations.

2.4.2 Reference: [Rohith, S., Jahnavi, L., Bhuvaneshwari, S.C., Supreeth, S. and Sujatha, B.K., 2020]

Summary: This paper discusses a use of logistic map-based encryption to secure medical image storage in the cloud environment. The authors propose the storage system based on a cloud with incorporation of the encryption technique for protecting patients' privacy. The challenge posed by data breaches, and unauthorized access in a secure cloud storage environment is described, along with proposed solutions for the security of medical images.

2.4.3 Reference: [Prabhavathi, K., Ananda Raju, M.B. and Ravi, V., 2023]

Summary: This paper addresses the idea of applying the logistic map-based encryption method for protection of medical images in wearable devices. This paper describes a lightweight scheme that may be used to encrypt content in resource-constrained

devices. It gives an energy efficiency and security performance of the proposed approach, establishing the usability of such a technique for wearable medical devices.

Logistic map-based image encryption has a wide number of applications in the medical field, such as:

i. Telemedicine: Secure transmission of medical images across telemedicine networks should be ensured to prevent exposure of the patient. Encryption using logistic map can be employed for encrypting the medical images before the act of transmission.
ii. Storage of Medical Images: Safe storage of medical images prevents access and data exposure by unauthorized persons. The storage of medical images should also be encrypted using the logistic map.
iii. Cloud-based medical image storage: Cloud storage provides several benefits, but at the same time, it puts security at a risk. Logistic map encryption for securing medical images stored in cloud.

2.5 ADVANCED TECHNIQUES AND FUTURE DIRECTIONS

2.5.1 Reference: [Kolivand, H., Hamood, S.F., Asadianfam, S. and Rahim, M.S., 2024]
Summary: This paper discusses using machine learning techniques in order to improve the security of a logistic map-based encryption scheme. The proposed approach in this paper adopts a machine learning-based automatic selection method for optimal logistic map parameters, thus achieving improved security and robustness of the encryption scheme.

2.5.2 Reference: [Abusukhon, A. and AlZu'bi, S., 2020]
Summary: The paper explores the use of quantum computing to break encryption based on logistic maps. The authors consider the possible implications of quantum computing for the security of classical encryption schemes and propose quantum-resistant encryption schemes.

2.5.3 Reference: [Kaur, J., Lamba, S. and Saini, P., 2021]

Summary: This paper introduces a hybrid encryption scheme combining the logistic map with a DNA-based encryption technique. The authors have simulated and experimented extensively to prove the enhanced security and robustness of the proposed scheme.

2.5.4 Reference: [Kaur, M., Singh, S. and Kaur, M., 2021]

Summary: This paper explores the application of chaos-based neural networks in image encryption. The authors propose a neural network-based encryption scheme that utilizes the chaotic properties of the logistic map. The paper demonstrates the effectiveness of the proposed scheme in terms of security and robustness.

2.5.5 Reference: [Zolfaghari, B. and Koshiba, T., 2022]

Summary: This paper explores the use of the logistic map-based encryption approach for protecting medical images within IoT settings. The paper presents a lightweight and power-efficient encryption scheme that could be implemented on IoT devices. It analyses the proposed scheme's security and performance in terms of latency, throughput, and energy usage.

2.5.6 Reference: [Xue, X., Zhou, D. and Zhou, C., 2020]

Summary: This paper discusses the use of blockchain technology to enhance the security of medical image sharing and storage. The authors have proposed a blockchain-based framework that incorporates logistic map-based encryption to protect the confidentiality and integrity of medical images. The paper discusses the advantages of using blockchain technology for secure medical image sharing and storage.

CHAPTER-3

METHODOLOGY

3.1 SYSTEM DESIGN AND IMPLEMENTATION

3.1.1 Image Acquisition:

Input will be in the form of medical images in standard formats, such as DICOM, JPEG, and PNG.

Images will be pre-processed to be compatible with the encryption algorithm.

3.1.2 Chaotic Key Generation:

The logistic map will be used to generate a chaotic sequence from an initial condition and a control parameter.

The generated chaotic sequence will be used as the encryption key.

3.1.3 Image Encryption:

- Pixel-Level Encryption: Each pixel of the image will be encrypted individually with a combination of chaotic sequence and proper encryption algorithm, such as XOR, or modular addition.
- Block-level Encryption:
  Dividing an image into pixels blocks. The block encrypts in the image, using t he block cipher, such as DES, or AES. It uses a chaotic sequence to produce the encryption key.

3.1.4 Image Decryption

The decryption process will involve the inverse operations of the encryption process, using the same chaotic key.

The encrypted image will be decrypted to recover the original image.

3.2 EVALUATION METRICS

To evaluate the performance of the proposed image encryption and decryption system, the following metrics will be used:

- Security Analysis:
- Key Sensitivity: The sensitivity of the encryption scheme to changes in the initial conditions of the logistic map will be analysed.

- Statistical Analysis: Histogram analysis and correlation analysis will be conducted on the encrypted images to test the randomness of such images.
- Resistance against Differential Attacks: The security of the encryption scheme will be tested regarding resistance against differential attacks.
- Key Space Analysis: A key space of the encryption scheme will be analysed against the strength of the given security.
- ➢ Computational Efficiency:
  The computational time taken for encryption and decryption will be measured to evaluate the efficiency of the algorithm.

3.3 EXPERIMENTAL SETUP

- Hardware: The experiments will be performed on a standard PC with a suitable processor and enough memory.
- Software: The system will be implemented in MATLAB, complemented by appropriate libraries of image processing and cryptography.

3.4 WORKING PROCEDURE

3.4.1 Pre-Steps involved in the working procedure

STEP-1: Obtaining the 1000 Logistic Sequences

Logistic Sequence generating code:

```
r = 3.9;

x0 = 0.1;

n = 1000;

x = zeros(1, n);

x(1) = x0;

for i = 2:n

   x(i) = r * x(i-1) * (1 - x(i-1));

end

disp('Generated sequence:');
```

```
disp(x);

figure;

plot(1:n, x, 'b.-');

title('Logistic Map Sequence');

xlabel('Iteration');

ylabel('x');
```

Code Explanation:

This MATLAB code generates and visualizes the behavior of the logistic map, a mathematical model used to study chaotic systems and population dynamics. Below is an explanation of the code step-by-step:

- Parameter Initialization:
  - r = 3.9: This is the growth rate parameter in the logistic map. It determines how chaotic or stable the system behaves.
  - x0 = 0.1: The initial value for the sequence, representing the starting point of the iteration.
  - n = 1000: This defines the number of iterations for the sequence.
- Array Setup:
  - x = zeros(1, n);: Creates an array of zeros with 1 row and n columns. This will store the sequence values.
  - x(1) = x0;: The first element of the sequence is set to the initial value, x0.
- Logistic Map Iteration:
  - The for loop iterates from 2 to n. Each iteration calculates the next value in the sequence using the logistic map formula:

\[

x_{i} = r \cdot x_{i-1} \cdot (1 - x_{i-1})

\]

Here:

- x_{i-1} is the value from the previous iteration.
- r is the growth rate.
- $(1 - x_{i-1})$ ensures the population is bounded between 0 and 1, mimicking constraints in a natural system.

- Display the Sequence:
- disp('Generated sequence:');: Prints the text "Generated sequence:".
- disp(x);: Displays the computed sequence values in the console.

- Visualization:
- figure;: Opens a new figure window for the plot.
- plot(1:n, x, 'b.-');: Plots the sequence, with the x-axis representing iteration numbers and the y-axis showing sequence values. The blue dots (b.-) represent the points, and lines connect them for visualization.
- title('Logistic Map Sequence');: Adds a title to the plot.
- xlabel('Iteration');: Labels the x-axis as "Iteration".
- ylabel('x');: Labels the y-axis as "x".

Purpose:

This code demonstrates how the logistic map evolves over time for a specific growth rate (r) and starting value (x0). It showcases key features of the system, such as fixed points, periodic orbits, or chaotic behavior, depending on the parameter r. Here, with $r = 3.9$, the system is chaotic.

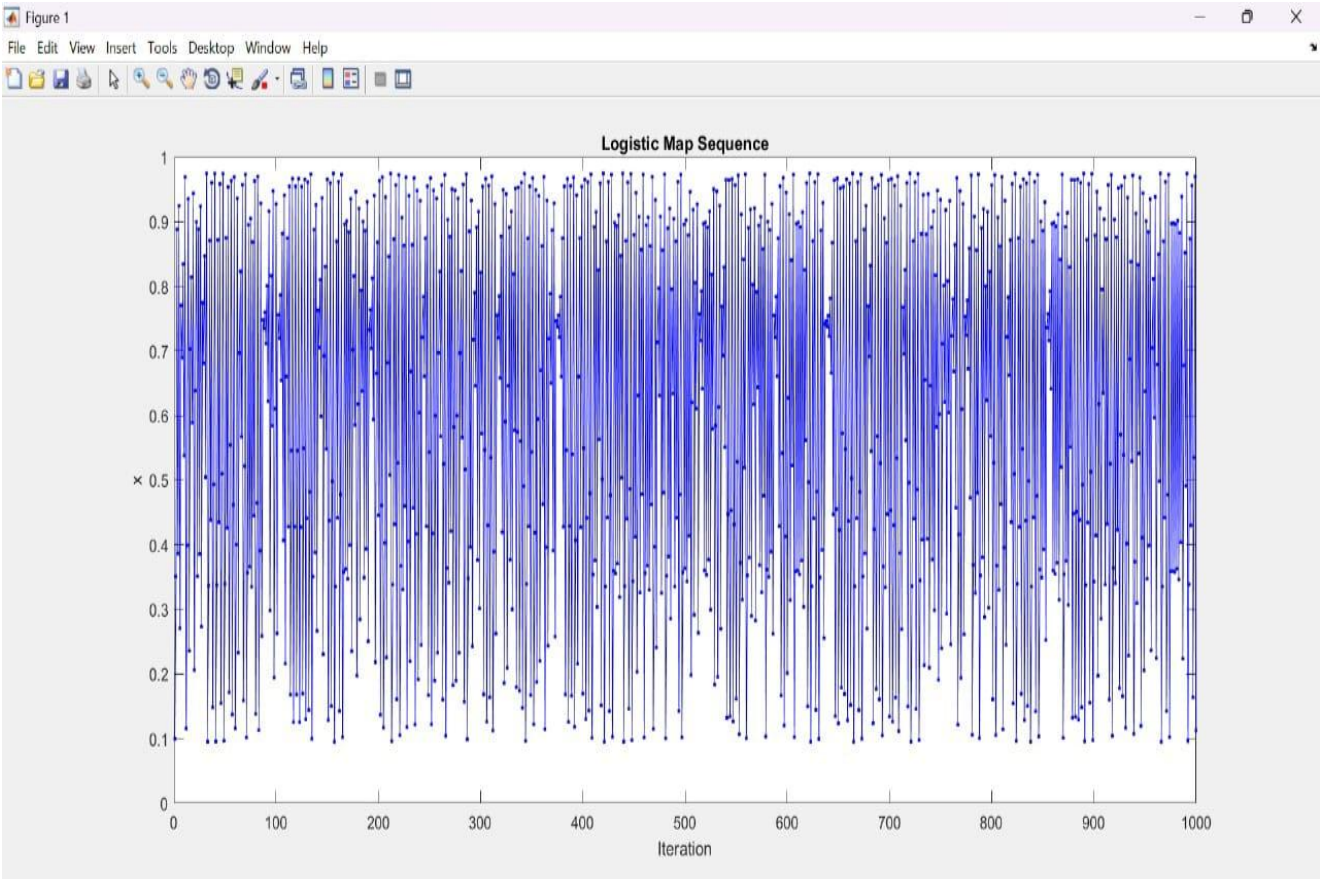


Fig 3.4.1 Logistic Map Sequence

STEP-2: Grayscale conversion

Basically this step is an optional in Medical Fields as the original image selected will be in grayscale which is basically from X-ray, CT scan , MRI scan. But conversion of colour pixeled images into grayscale image helps in better encryption. So it is an essential step if we consider our input image as colour pixeled images.

Gray scaling an image in the context of image encryption, especially when using a logistic map, involves converting the original image into shades of gray and then applying encryption techniques to ensure security. Here's a detailed explanation:

Original Image Representation: A digital image is typically represented in RGB (Red, Green, Blue) format, where each pixel has three colour intensity values ranging from 0 to 255.

Conversion to Gray Scale:

Each pixel in the RGB image is converted to a single intensity value (grayscale value) using a weighted formula. A common formula is:

Gray Value = 0.2989 × R+ 0.5870 × G + 0.1140 × B

0.2989 – It is the weight assigned to the red component, it is a constant value. It reflects the human eye's sensitivity to red light. Human eye is less sensitive to red colour compared to green so the component value is less.

R – It is the red channel value of the colour, ranges from 0 to 255 in 8bit RGB colour model.

0.5870 – It is the weight assigned to green component. As Human eye is more sensitive to green colour the component value is comparatively higher than red colour.

G – The green channel value of the colour, it also ranges from 0 to 255.

0.1140 – It is the weight assigned to blue component. As human eye has least sensitivity towards blue colour it has the least value int the equation.

B – Blue channel value of colour, it also ranges from 0 to 255.

This ensures that the human eye's sensitivity to different colours is accounted for, giving a more perceptually accurate gray image.

The below image is an example of conversion of colour pixeled image to grayscale image.

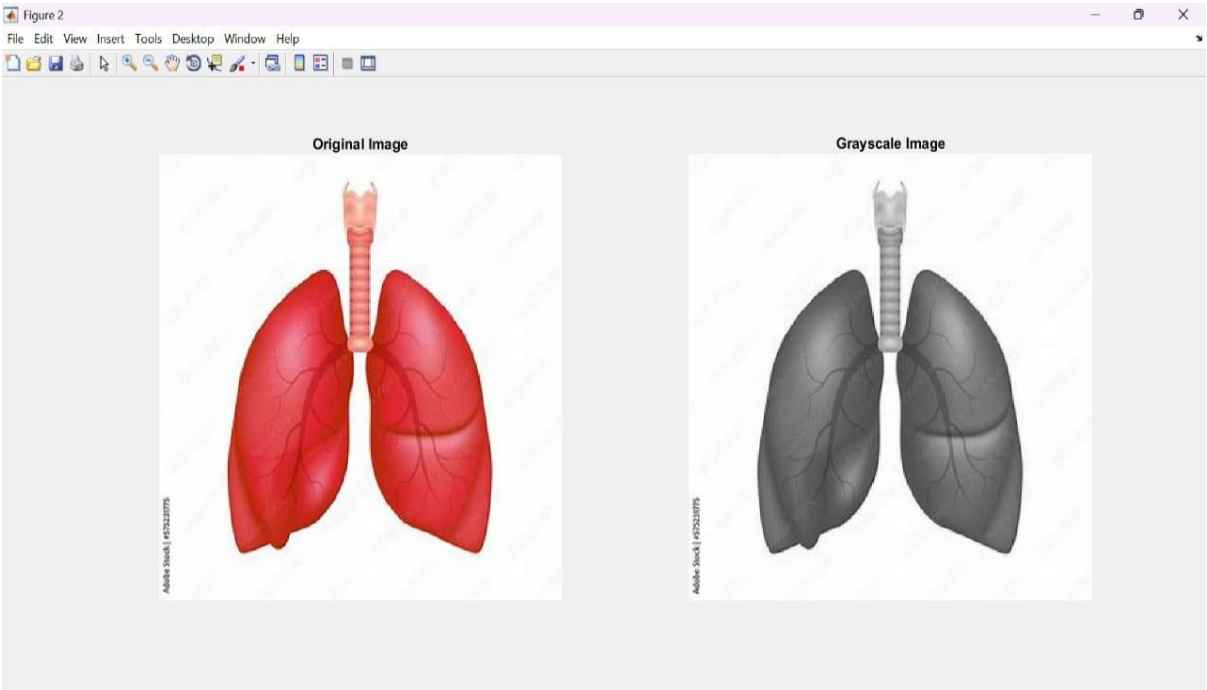


Fig 3.4.2 Original image to Grayscale image conversion

3.4.2 Steps involved in Working procedure

- Introduction to the Logistic Map

The logistic map is a mathematical equation used to generate chaotic sequences. It's represented as:

$$x_{n+1} = r \cdot x_n \cdot (1 - x_n)$$

- Here, x_n is a value between 0 and 1, and r (a constant) determines the behavior of the sequence.

- For certain values of r (e.g., 3.9), the logistic map produces highly unpredictable, chaotic sequences.

- Image Encryption Process

The encryption process involves several steps to transform the original image into an encrypted, unreadable form.

Step 1: Convert the Image into Pixel Data

- The image is first loaded into memory and broken down into its individual pixels.
- Each pixel is represented by numerical values, such as grayscale intensity (0–255) or RGB values.

Step 2: Generate the Chaotic Sequence

- Using the logistic map, a chaotic sequence is generated.
- An initial value x_0 (called the seed) and the parameter r are chosen. These act as keys for the encryption process.
- The chaotic sequence is scaled to match the range of pixel values (e.g., 0–255).

Step 3: Shuffle the Pixel Data

- The chaotic sequence is used to rearrange the positions of pixels in the image. This step ensures that the original structure of the image is scrambled.

Step 4: Modify Pixel Intensity

- Another chaotic sequence is used to alter the intensity of the pixels. This is usually done using operations like XOR, addition, or subtraction.
- For example: $\text{Encrypted Pixel} = (\text{Original Pixel} \oplus \text{Chaotic Value}) \mod 256$.

Step 5: Output the Encrypted Image

- The scrambled and intensity-modified pixel values are saved as a new image. This encrypted image appears as noise and is incomprehensible.

- Image Decryption Process

The decryption process reverses the encryption steps using the same logistic map parameters and keys.

Step 1: Load the Encrypted Image

- The encrypted image is read, and its pixel values are extracted.

Step 2: Regenerate the Chaotic Sequence

- The same logistic map equation, initial value x_0, and parameter r are used to regenerate the chaotic sequence.

Step 3: Reverse Pixel Intensity Modification

- The pixel intensity modification step is reversed using the same operation (e.g., XOR with the chaotic value).

Step 4: Undo the Pixel Shuffling

- The chaotic sequence is used to determine the original positions of the pixels, reversing the shuffling.

Step 5: Reconstruct the Image

- After reversing both the shuffling and intensity modification, the original image is reconstructed.

- Key Points
  - Initial Parameters (Seed and \(r\)): These act as keys and must be known for successful decryption.
  - Sensitivity: Even a slight change in the initial value or parameters will result in a completely different output, making the method highly secure.
  - Efficiency: The process is computationally efficient, suitable for real-time applications.
- Applications

Logistic map-based encryption is widely used in secure image transmission, where confidentiality and protection against unauthorized access are critical.

This explanation captures the essence of image encryption and decryption using logistic maps in a clear, straightforward manner. Let me know if you'd like a deeper dive into the mathematics or code examples!

CHAPTER-4

BLOCK DIAGRAM

4.1 Domain types in Image System

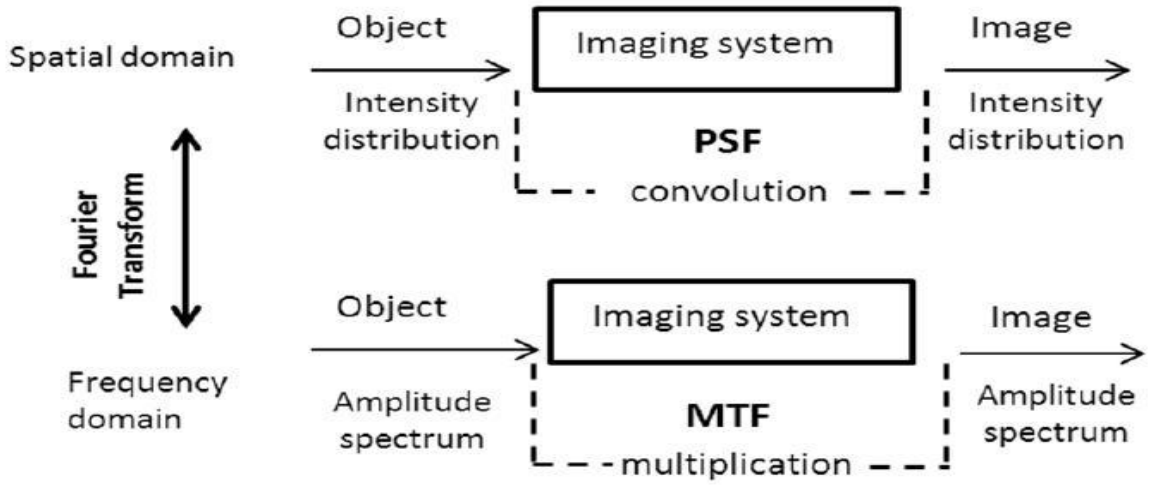


Fig 4.1.1 Image system Representing in Spatial and Frequency Domain

4.2 Image Encryption and Decryption process

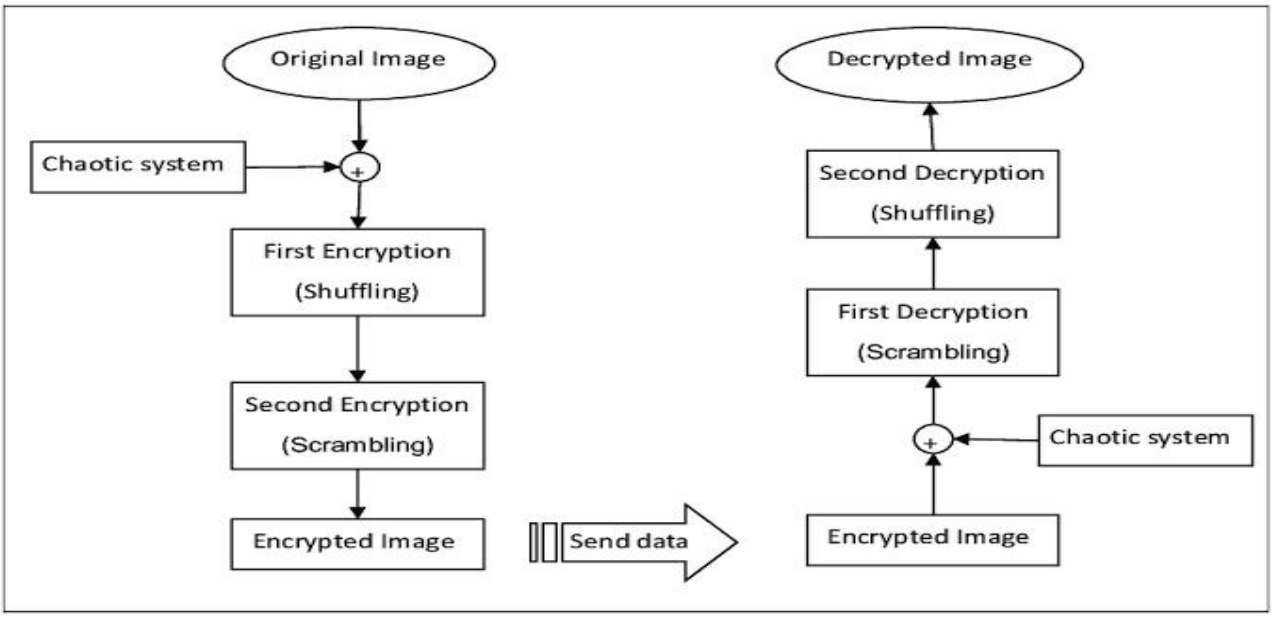


Fig 4.2.1 Block Diagram representing the process of encryption and decryption of image

4.1.1 Explanation of Block Diagram

Here is an explanation of the steps shown in the diagram:

Encryption Process:

- Original Image: The process starts with an original image that needs to be encrypted.
- Chaotic System: A chaotic system produces a pseudo-random sequence or key that is added to the original image. This step is usually performed to introduce randomness and enhance security.
- First Encryption (Shuffling): The image, combined with the chaotic system output, undergoes the first encryption stage, which is shuffling the pixel positions. This step rearranges the pixel positions to obscure the image.
- Second Encryption (Scrambling): The shuffled image is passed through a second encryption stage, which scrambles the pixel values further. This step alters the pixel values to further encrypt the image.
- Image: After the second stage of encryption, the final encrypted image will result. This encrypted image will now be sent to the receiver.

Process for decryption:

- Receipt of Encrypted Image: After encrypting the image, this encrypted image reaches the receiver.
- First decryption scrambling: The first stage of decryption process will be reversing all scrambling of pixel values in that particular image done at the time of the second encryption.
- Second Decryption (Shuffling): After the first decryption stage of the image, the second decryption of the same image is carried out by undoing the shuffling of the pixel positions, that is, done in the encryption stage.
- Chaotic System: The same pseudo-random sequence or key of the chaotic system, that has been used in the process of encryption, is subtracted from the decrypted image to find the original pixel values.
- Decrypted Image: The result of the second stage of decryption combined with the output of the chaotic system gives a decrypted image that should be exactly the same as the original image.

CHAPTER-5

SOFTWARE (MATLAB: A POWERFUL TOOL FOR IMAGE PROCESSING)

5.1 MATLAB

MATLAB is a versatile language and environment that is widely applied in scientific and engineering applications, including image processing. Its built-in functions and toolboxes, along with its intuitive interface, have made it the ideal choice for image analysis and manipulation. Key Features of MATLAB in Image Processing.

- Image Acquisition and Display:
  - imread: Reads images from multiple formats, such as JPEG, PNG etc.,
  - imshow: Displays images.
- Image Preprocessing:
  - imresize: Resizes images.
  - imcrop: Crop images.
  - imrotate: Rotate images.
- Image Enhancement:
  - imadjust: Adjust image contrast.
  - histeq: Histogram equalization.
  - medfilt2: Median filtering for noise reduction.
- Image Filtering:
  - imfilter: Linear and nonlinear filters.
  - fspecial: Filter kernels (e.g., Gaussian, Laplacian).
- Image Segmentation:
  - imsegkmeans: K-means clustering for segmentation.
  - imbinarize: Threshold images.
- Feature Extraction:
  - edge: Edge detection in images.
  - graycoprops: Texture features.

- Image Analysis:
  - imhist: Plot image histograms.
  - regionprops: Properties of image regions.

Advantages of Using MATLAB for Image Processing:
It also offers a very user-friendly environment when dealing with images. MATLAB has plenty of toolboxes. The most important and wide toolbox is Image Processing Toolbox with the set of functions provided for various image processing operations. Its capabilities in displaying images along with the results have been used in the code. Because MATLAB uses scripting language, prototyping comes very easily and fast; it facilitates fast experimentation, and so its integration with other tools like Python and C++ languages.

5.2 SOFTWARE STIMULATION

5.2.1 Main Code

```
clear all;

clc;

image = imread('C:\Program Files\MATLAB\R2017b\toolbox\images\imdata\lungs.jpeg');

if size(image, 3) == 3

   image1 = rgb2gray(image); % Convert to grayscale if image is RGB

else

   image1 = image;

end

image1 = imresize(image1, [256 256]);

[m, n] = size(image1);

subplot(2, 2, 1), imshow(image1);

title('Original Image');

x = 0.1;

r = 3.99;
```

```
seq = zeros(1, m * n);
for k = 1:(m * n - 1)
    x = r * x * (1 - x);
    seq(k) = x * 255;
end
seq = round(seq);
image4 = zeros(m, n, 'uint8');
image5 = zeros(m, n, 'uint8');
count = 1;
for i = 1:m
    for j = 1:n
        image4(i, j) = bitxor(seq(count), image1(i, j));
        image5(i, j) = bitxor(seq(count), image4(i, j));
        count = count + 1;
    end
end
subplot(2, 2, 2), imshow(image4);
title('Encrypted Image');
subplot(2, 2, 3), imshow(image5);
title('Decrypted Image');
subplot(2, 2, 4), imhist(image4);
title('Histogram of Encrypted Image');
j = entropy(image4);
disp(['Entropy of Encrypted Image: ', num2str(j)]);
```

5.2.2 Code Explanation

This MATLAB code demonstrates a basic encryption and decryption process for an image using a chaotic sequence generated by the logistic map. Here's a breakdown of the code:

- Image Loading and Preprocessing

- The image is read from a file path using imread.

- If the image is in color (RGB), it is converted to grayscale using rgb2gray.

- The grayscale image is resized to a fixed size of 256x256 using imresize for uniformity.

- The dimensions (m and n) of the resized image are obtained, and the original image is displayed using imshow.

- Generating a Chaotic Sequence

- The logistic map, a simple mathematical model for chaotic behavior, is used to create a pseudo-random sequence.

- The initial value x and the parameter r determine the chaotic sequence's characteristics.

- The sequence is generated by iterating the logistic equation:

$x_{k+1} = r \cdot x_k \cdot (1 - x_k)$,

where the sequence values are scaled to lie between 0 and 255 (suitable for image processing).

- Encryption and Decryption
  - Two matrices (image4 and image5) are preallocated to store the encrypted and decrypted images.
  - For encryption:
  - Each pixel value of the image is XOR-ed with a corresponding value from the chaotic sequence, ensuring a unique transformation.
  - For decryption:
  - The XOR operation is applied again with the same chaotic sequence, which reverses the encryption.

- A counter (count) keeps track of the position in the chaotic sequence for each pixel.

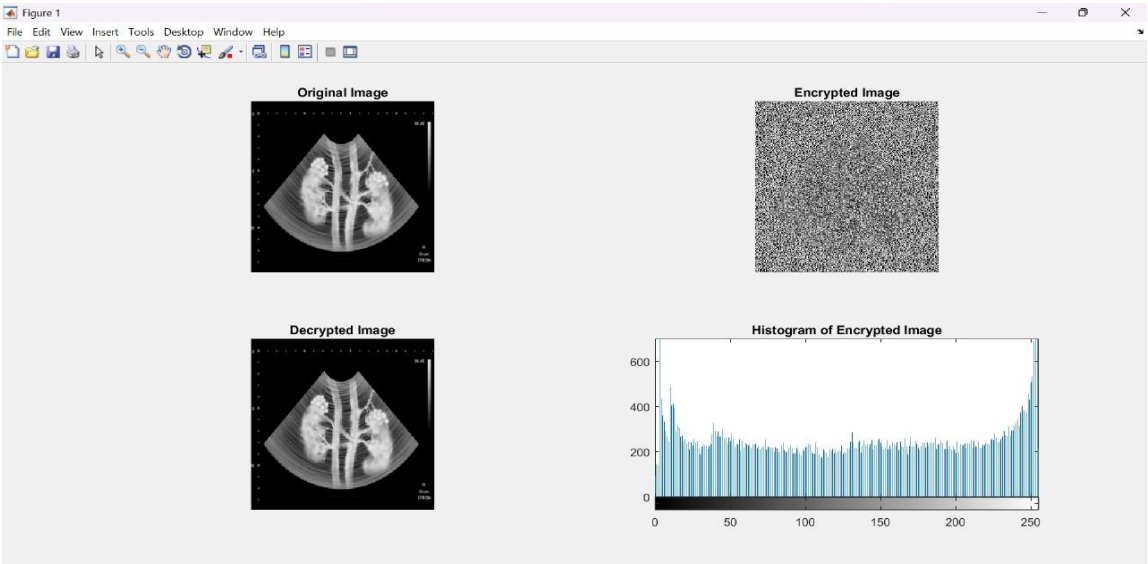
- Displaying Results
- The original image, encrypted image, decrypted image, and the histogram of the encrypted image are displayed using subplot.
- The histogram of the encrypted image is typically more uniform, indicating effective encryption by spreading pixel values.

- Entropy Calculation
- The entropy of the encrypted image is computed using entropy. Higher entropy indicates better randomness and security in the encryption process.

- Summary of Outputs:
- Original Image: The pre processed grayscale image.
- Encrypted Image: The image after applying chaotic encryption.
- Decrypted Image: The reconstructed image from decryption, which matches the original image.
- Histogram: Shows the pixel value distribution of the encrypted image.
- Entropy: Quantifies the randomness of the encrypted image; higher values indicate better encryption security.

This process illustrates a simple method for image encryption using chaos theory, where the logistic map plays a critical role in generating a secure key.

Result and Discussion



The logistic map, a mathematical function derived from chaos theory, serves as the key ingredient for this transformation. It generates a sequence of seemingly random numbers based on an initial value (seed) and a defined equation. These numbers guide how the pixels in the image are rearranged or modified.

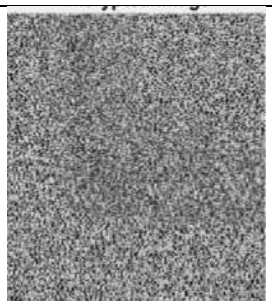
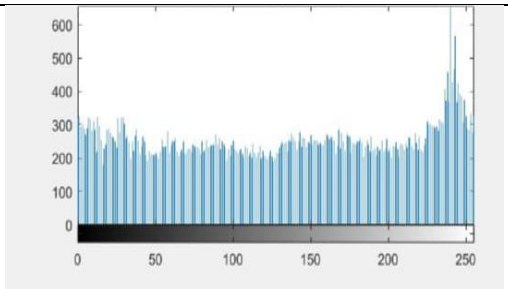
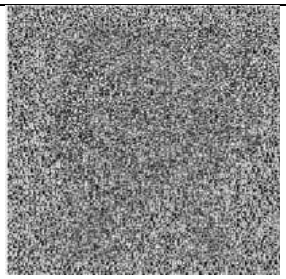
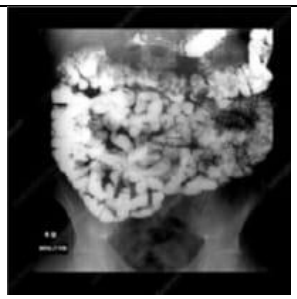
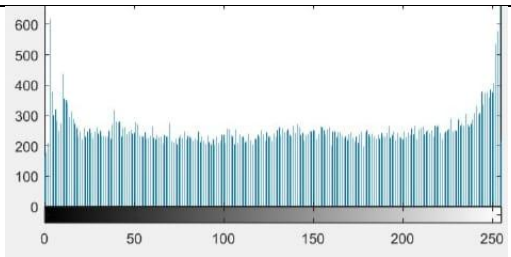
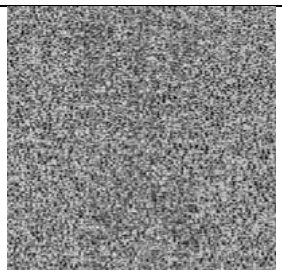
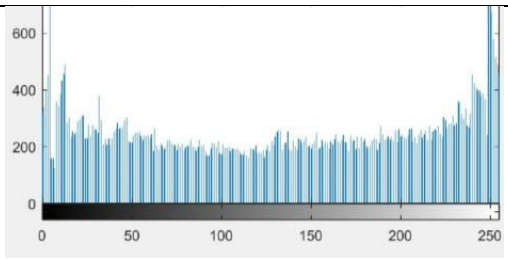
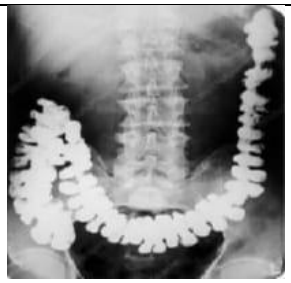
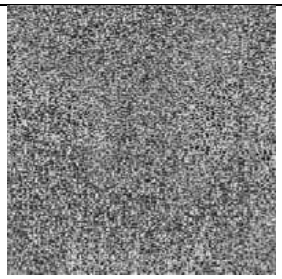
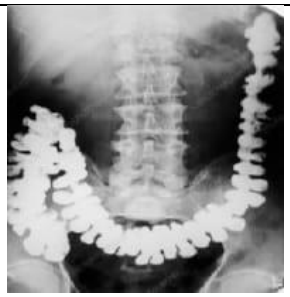
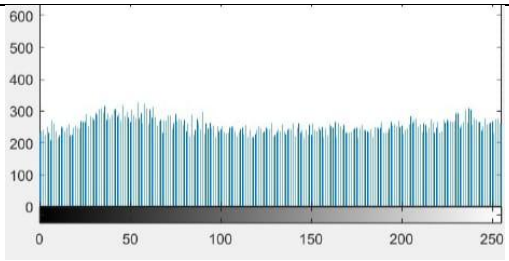
During Encryption:

The logistic map generates a sequence that dictates which pixels to shuffle or alter and in what order.This results in a completely disordered image, making it impossible to deduce the original visual content without knowing the exact logistic map parameters (key).

Decryption:

The same logistic map parameters (seed and equation) are used to reverse the scrambling process.This reorders the pixels back to their original positions and restores their original values, recreating the original image.

The result of this encryption and decryption process demonstrates the effectiveness of chaotic systems like the logistic map in ensuring both security and recoverability. The encrypted image appears random and secure, while the decryption process perfectly restores the original image, provided the correct parameters are used. If the decryption key is incorrect or slightly altered, the original image cannot be retrieved, showcasing the sensitivity and robustness of the method.

SL No	ORIGINAL IMAGE	ENCRYPTED IMAGE	DECRYPTED IMAGE	HISTOGRAM OF ORIGINAL IMAGE
1.				
2.				
3.				
4.				

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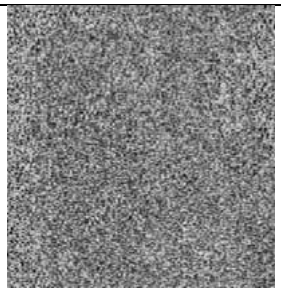
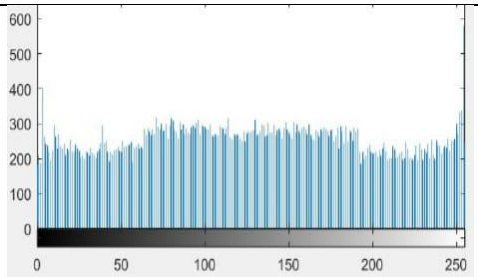
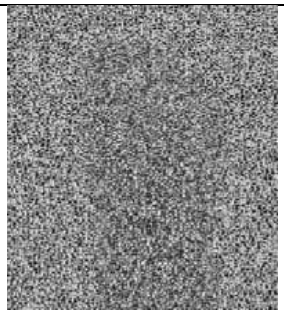
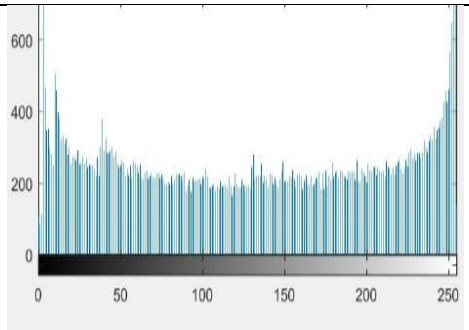
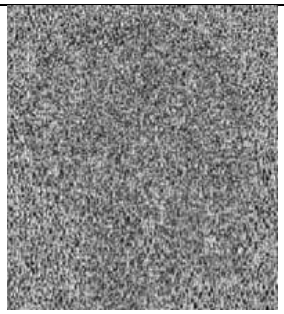
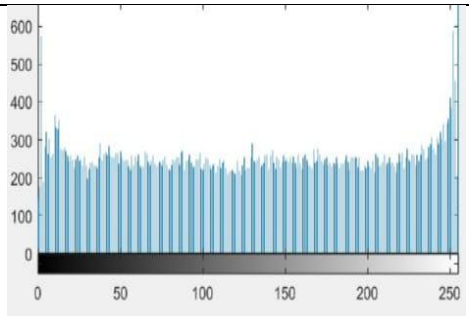
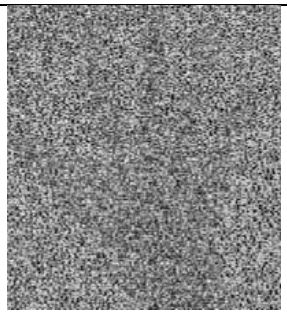
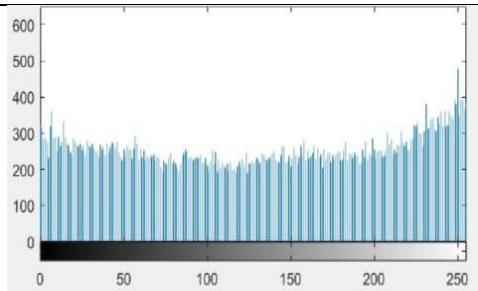
5.				
6.				
7.				
8.				

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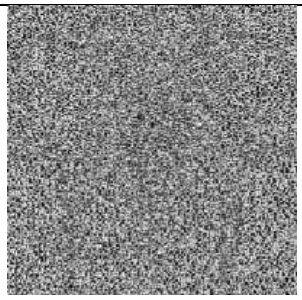
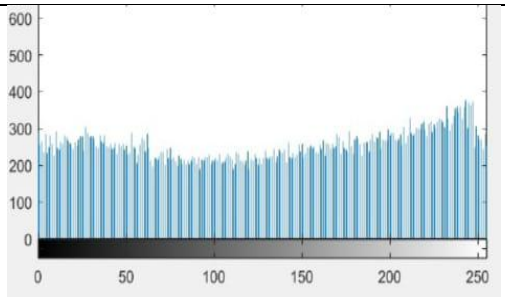
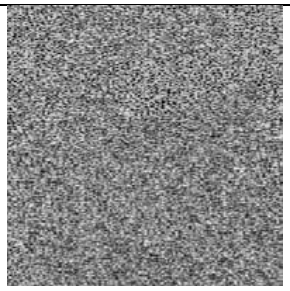
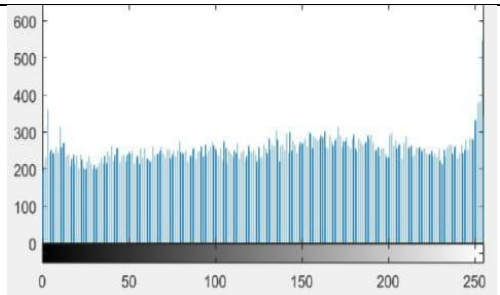
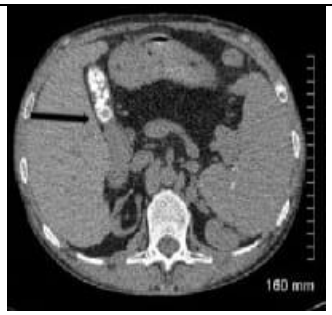
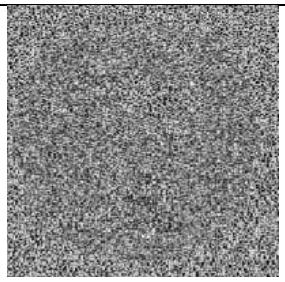
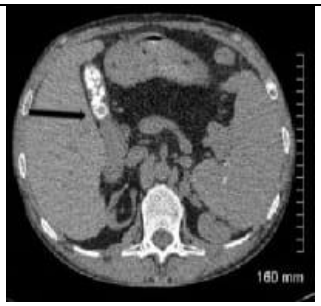
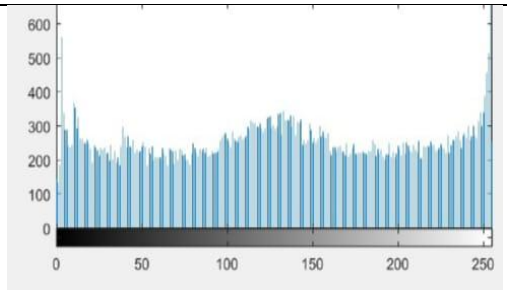
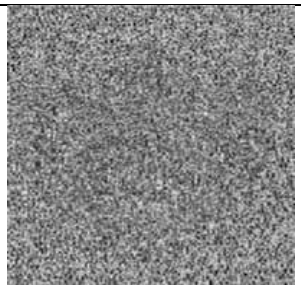
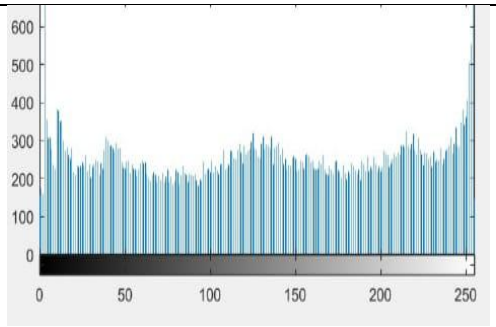
9.				
10.				
11.				
12.				

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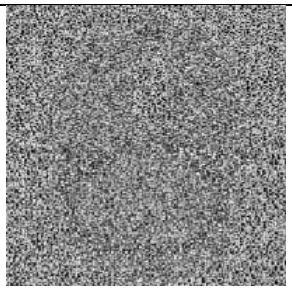
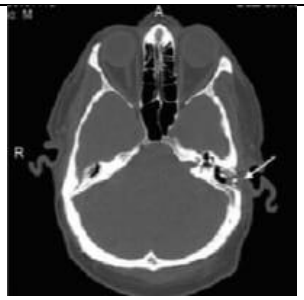
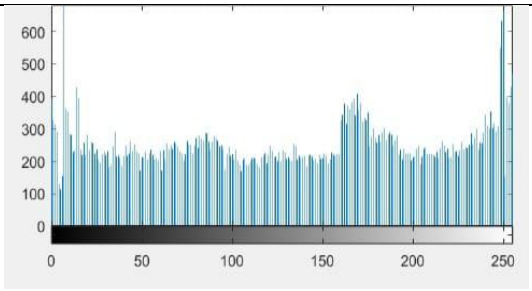
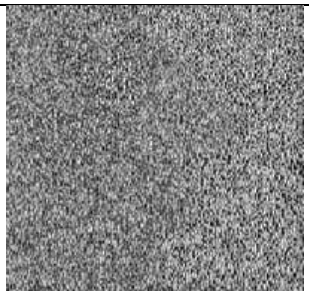
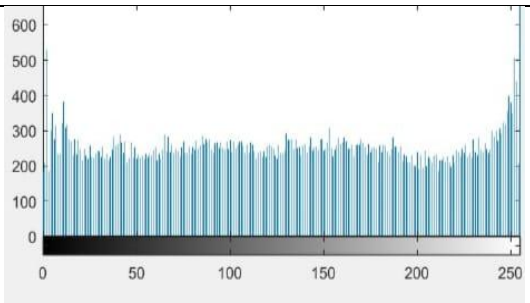
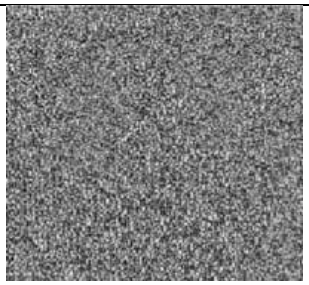
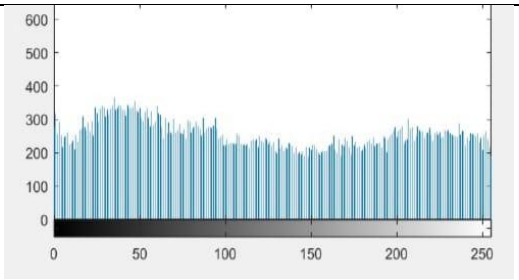
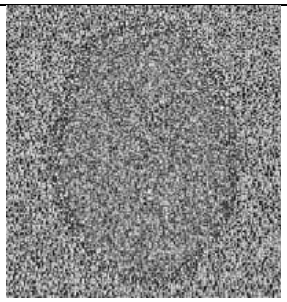
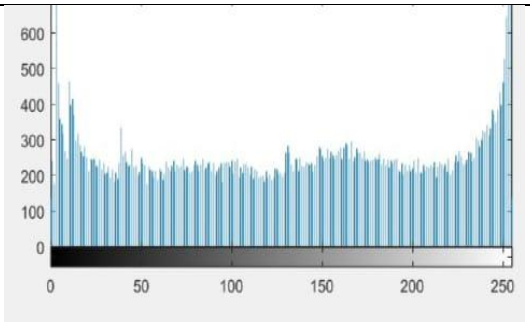
13.				
14.				
15.				
16.				

Future Scope

Although logistic map-based image encryption has shown promising results, there are several avenues for future research and development:

- Improving Security
  - Quantum-Resilient Encryption: Since quantum computation is growing fast, encryption schemes have to be prepared to face the new challenges against quantum attacks. Hybrid post-quantum cryptography schemes, by using problems difficult to be solved even by quantum computers, can be applied in the logistic map-based approach.
  - Hybrid Encryption Schemes: Hybrid scheme can be considered for combining techniques based on the logistic map for encryption purposes along with the other techniques available, block ciphers, or even public-key cryptography.
  - Adaptive Security: Designing encryption algorithms that adapt to the changing security threats and the changing techniques of attacks.
- Efficiency Enhancement
  - Hardware Acceleration: Implementation of encryption and decryption algorithms on dedicated hardware such as FPGAs or GPUs significantly improves the computational efficiency.
  - Parallel Processing: Applying parallel processing techniques to speed up the encryption and decryption processes, especially for large medical images.
  - Optimized Implementations: Optimizing the implementation of the logistic map and encryption algorithms to reduce the computational overhead.
- Specific Medical Image Challenges Addressment
  - 3D Medical Image Encryption: Developing encryption techniques that can effectively handle 3D medical images, such as CT scans and MRIs.
  - Secure Cloud Storage: Ensuring the security of medical images stored in cloud environments with the integration of robust encryption techniques.
  - Real-time Encryption: Developing effective encryption algorithms that can be utilized in real-time medical imaging applications, such as telemedicine.
- Interdisciplinary Approaches

- Bio-inspired Encryption: Study of bio-inspired encryption techniques, such as DNA-based encryption, to enhance the security of medical images.
- Machine Learning-Based Encryption: Use of machine learning techniques to automatically adapt the encryption parameters to the specific characteristics of medical images.